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Early Life Stress, DNA Methylation of *NR3C1* and *HSD11B2*, and Oral Feeding Skill Development in Preterm Infants:

A Pilot Study

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Abstract

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The authors declare no conflicts of interest.

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Ethical approval was provided by the Institutional Review Board of the Loyola University Chicago in advance of implementation.

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Background: Early life stress exposure in preterm infants may alter DNA methylation of *NR3C1* and *HSD11B2*, disrupting neurobehaviors needed for oral feeding (PO) skill development.

Purpose: To (1) examine the feasibility of the study protocol; (2) describe early life stress, DNA methylation of *NR3C1* and *HSD11B2*, and PO skill development; and (3) explore the association between DNA methylation of *NR3C1* and *HSD11B2* and infant characteristics, early life stress, and PO skill development.

Method: We employed a longitudinal descriptive pilot study (N = 10). Infant characteristics were collected from the infant's electronic medical record. Early life stress was assessed via the modified Neonatal Infant Stressor Scale. DNA methylation of *NR3C1* exon 1F and *HSD11B2* promoter regions was analyzed from the infant's buccal samples. PO skill development was evaluated using the Early Feeding Skills Assessment.

Results: Infants who experienced more acute and chronic stress during their neonatal intensive care unit hospitalization demonstrated higher DNA methylation at CpG 17 and 31 of the *NR3C1* exon 1F and at CpG 4 and 28 of the *HSD11B2* promoter regions. Infants with higher DNA methylation at these CpG sites also exhibited less optimal PO skill development and experienced longer transition from first to full PO.

Implications for Practice and Research: Our findings revealed relationships among early life stress, DNA methylation of *NR3C1* and *HSD11B2*, and PO skill development in preterm infants. Future research is warranted to examine the multiomics pathways whereby early life stress influences the phenotypes of infant outcomes.

Keywords

DNA methylation; early life stress; epigenetics; glucocorticoid genes; *HSD11B2*; NICU; *NR3C1*; oral feeding skills

BACKGROUND AND SIGNIFICANCE

Preterm infants admitted to the neonatal intensive care unit (NICU) are exposed to excessive early life stress.¹ Exposure to early life stress results in epigenetic modifications of glucocorticoid-related genes, ie, DNA methylation of the glucocorticoid receptor gene (*NR3C1*) and the 11 β -hydroxysteroid dehydrogenase type 2 gene (*HSD11B2*).^{1,2} DNA methylation of *NR3C1* and *HSD11B2* may alter the hypothalamic-pituitary-adrenal (HPA) axis regulation of cortisol.²⁻⁴ This may interfere with the process of neurobehavioral development and lead to poor neonatal outcomes,^{2,5-7} ie, delayed oral feeding (PO) skill development. Delayed achievement of PO skill development may lead to failure to thrive, prolonged hospitalization, and increased healthcare cost.⁸ There is a critical need to clarify the extent to which epigenetic modifications due to early life stress influence PO skill development. Findings from the study are important for neonatal nurses as they will inform clinical practice and facilitate understanding toward a clinically modifiable mechanism that may impact preterm infants.

LITERATURE REVIEW

Preterm infants experience early life stress due to the NICU environment, health conditions, caregiving interventions, and prolonged parental separation.^{4,7} Early life stress may exceed the infant's regulatory capacity and alter the allostatic balance.⁹ A large body of evidence demonstrates that early life adversities and environmental stimuli can lead to significant epigenetic modifications, particularly in the genes *NR3C1* and *HSD11B2*, which are crucial for the stress response system.^{2,3,7,10–12} Epigenetics is a field of study that focuses on phenotypic changes that do not involve alterations in the DNA sequence.² DNA methylation is one of the most commonly recognized epigenetic modifications. DNA encodes genes using a sequence of nucleotide bases—adenine, thymine, guanine, and cytosine.^{2,8} DNA methylation is a biochemical process in which methyl groups are attached to cytosine.^{2,8} DNA methylation affects gene expression levels, functioning like a dimmer switch that can either increase or decrease expression to impact phenotypic outcomes.^{2,8} The *NR3C1* gene encodes the glucocorticoid receptors, essential for regulating the HPA axis. In preterm infants, early life stress can result in increased DNA methylation of *NR3C1*, potentially reducing glucocorticoid receptor expression and impairing stress regulation.⁶ Similarly, the *HSD11B2* gene encodes the enzyme 11 β -hydroxysteroid dehydrogenase type 2, which inactivates glucocorticoids.⁶ Early life stress can lead to increased DNA methylation of *HSD11B2*, decreasing enzyme levels and causing prolonged exposure to active glucocorticoids, thereby increasing stress sensitivity.^{2,13}

DNA methylation of *NR3C1* and *HSD11B2* regulates the body's key glucocorticoid homeostasis, ie, cortisol.⁸ Preterm infants exposed to high levels of stress show altered cortisol reactivity, which can have long-term implications for their physical and psychological health.¹⁴ Excess cortisol at critical windows of development can disrupt the process of neurobehavioral development.^{2–7,15–17} Evidence demonstrates that the integrity of neurobehavioral development is crucial for PO skill development.^{18–21} It is possible that excessive and prolonged early life stress may alter DNA methylation of *NR3C1* and *HSD11B2*. This in turn can result in cortisol dysregulation and subsequent impaired neurobehavioral development and delayed PO skill development. There is limited understanding regarding the complex relationships among early life stress, DNA methylation of *NR3C1* and *HSD11B2*, and PO skill development in preterm infants. The aims of this pilot study were to (1) examine the feasibility of the study protocol; (2) describe early life stress, DNA methylation of *NR3C1* exon 1F and *HSD11B2*, and PO skill development in preterm infants during NICU hospitalization; and (3) explore the association between DNA methylation of *NR3C1* exon 1F and *HSD11B2* and infant characteristics, early life stress, and PO skill development.

METHODS

Design and Setting

This pilot study employed a longitudinal descriptive design. It was conducted at a level III open-bay NICU. Infants received the study site's standard of care, eg, developmental and clustered care. Infants from our study site are primarily bottle-fed during NICU

hospitalization. The Loyola University Chicago Institutional Review Board approved the study.

Sample

We enrolled a sample of 11 preterm infants in the study. Inclusion criteria were birth between 26 and 34 weeks gestational age (GA). Exclusion criteria were as follows: necrotizing enterocolitis, gastrointestinal defects, oral cavity defects, intraventricular hemorrhage grade III or IV, periventricular leukomalacia, chromosomal abnormalities, cardiac surgery, received glucocorticoid therapy during NICU hospitalization, or unresolved oral ventilation, sepsis or pneumonia by 34 weeks postmenstrual age (PMA). We selected these exclusion criteria to reduce confounding by variables unrelated to early life stress that may interfere with PO skill development and cortisol levels. We excluded preterm infants older than 34 weeks GA because their NICU hospitalization is generally brief and would not allow us sufficient time to examine NICU early life stress.

MEASURES

Infant Characteristics

A research nurse reviewed the infant's electronic medical record to extract the following: infant's sex, race, ethnicity, GA (weeks), birth weight (g), delivery type (vaginal/C-section), Apgar score (1-minute and 5-minute), small for GA (yes/no), respiratory distress syndrome (yes/no), health status during NICU hospitalization via the Neonatal Medical Index,²² duration of exclusive tube feedings (number of days infants received only gavage feedings, no PO transition), duration of tube feeding (number of days infants receive gavage feedings, including PO transition), duration of transition from first to full PO, duration of NICU hospitalization, and PMA (weeks) at first PO, full PO, and discharge. Day of first PO was defined as the first of at least 2 consecutive days when the infant was able to orally consume $\geq 10\%$ of their prescribed feedings.²³ Day of full PO was defined as the first of at least 2 consecutive days when the infant was able to orally consume 100% of their prescribed feedings.²³ The Neonatal Medical Index summarizes infants' medical condition with classifications ranging from 1 for infants born weighing ≥ 1000 g and without major complications to 5 for infants born weighing < 1000 g and with very serious complications.²²

Early Life Stress

A research nurse assessed the infant's exposure to early life stress daily during the entire NICU hospitalization using data from electronic medical record. The modified Neonatal Infant Stressor Scale (NISS) assesses 47 acute stress events (eg, diaper changes, x-ray, and intravenous injection) and 23 chronic stress events (eg, intranasal oxygen and nothing by mouth (NPO)), as described previously.²⁴ The acute and chronic stress events were categorized into 4 severity levels (2 = little, 3 = moderate, 4 = very, and 5 = extremely). Acute stress events in the second and third categories were considered the expected standard of care in the NICU, while the fourth and fifth categories were considered "very" and "extreme." The researchers were not blinded to early life stress data. Predictive and construct validity have been established, ie, correlations between higher stress scores and brain structural changes, poorer neurobehaviors, and increased cortisol.²⁵ Acute stress was

examined based on the frequency of events, and chronic stress was examined based on the number of hours of events. For each infant, the unweighted daily acute stress and unweighted daily chronic stress were calculated by summing the frequency of acute stress events and number of hours for chronic stress events that the infant experienced on a particular day, respectively. The mean daily acute stress and mean daily chronic stress were then computed from these individual unweighted daily values. The total acute stress and total chronic stress were computed by adding all of the frequencies for acute stress events and the number of hours for chronic stress events over the entire NICU hospitalization, respectively.

Oral Feeding Skill Development

The Principal Investigator (PI) or speech therapist on the research team assessed PO skill development during a feeding session, in-person, at the bedside, once a week after the first PO during the infant's NICU hospitalization. Infants were fed by either the PI or speech therapist. PO skill development was assessed using the Early Feeding Skills Assessment (EFS).^{26,27} EFS includes 22 items across 5 subscales: respiratory regulation, oral-motor functioning, swallowing coordination, engagement, and physiologic stability. EFS skill items were scored using the following 3-option structure: not yet evident, emerging, or consistently observed. Indicators of problems within a skill were scored using the following 3-option structure: frequently observed, occasionally observed, or never or rarely observed. Each item received a score of 1, 2, or 3. A score of 1 represented the least skill or highest frequency of problems. A score of 2 represented skills that were still emerging or occasionally observed or problems were occasionally observed. A score of 3 represented a mature skill or absence of problems. Scores for each subscale and the total summary were calculated. The total summary score ranged from 22 to 66. A higher subscale score and a higher total summary score were considered optimal. Inter-rater reliability and construct validity have been established, ie, Cronbach $\alpha = 0.81$, with correlations between EFS scores and GA and PMA. The speech therapist was trained by the PI to score the EFS. They also conducted 30% of the in-person EFS assessments together and discussed any discrepancies until 100% agreement was achieved.

DNA Methylation of *NR3C1* and *HSD11B2*

The DNA methylation of *NR3C1* exon 1F and *HSD11B2* promoter regions were sequenced from the infant's buccal cell DNA.⁷ At 34 weeks PMA, one buccal swab sample from each infant using Isohelix DNA Buccal Swabs (Boca Scientific) was collected. The buccal sample was collected anytime between 34 weeks 0 days PMA and 34 weeks 6 days PMA. If the infant was born at 34 weeks GA, we waited until 34 weeks 6 days PMA to collect the sample. This time was selected because it was an ideal time to allow for assessment of NICU early life stress, and it is also around the time infants at our study site would start PO. Samples were obtained immediately before feeding or at least 30 minutes after feeding to avoid contamination and frozen at -70°C until analysis.

Genomic DNA was extracted from the buccal swab samples using a Maxwell RSC Buccal Swab DNA kit (Promega) implemented on an automated Maxwell RSC48 device. The DNA methylation for *NR3C1* exon 1F (31 CpG sites) and *HSD11B2* (28 CpG

sites) promoter regions was assessed using methods described previously,²⁸ such as deep amplicon sequencing from bisulfite-converted DNA using next-generation sequencing (see Supplemental Digital Content, Supplemental Description for DNA Methylation Analysis, available at: <http://links.lww.com/ANC/A295>). Raw sequence data were imported into the software package CLC Genomics Workbench (v22; Qiagen) for analysis. The percent methylation at CpG sites was determined by the ratio of C residues (derived from 5-methylcytosine) to C + T residues (derived from 5-methylcytosine plus cytosine, respectively) at any site within the DNA sequence of *NR3C1* exon 1F and *HSD11B2* promoter regions. In addition, the DNA methylation status of each read was assessed individually by counting the number of methylated CpG sites observed within that read and summed to assess average methylation status in each amplicon.

Procedure

Upon NICU admission, we screened for eligible infants using the inclusion and exclusion criteria. If infants were eligible, the research nurse approached their parents for consent within 3 days after birth. Written informed consent was obtained from parents. After obtaining the informed consent, data pertaining to infant characteristics and daily early life stress were collected from the infant's electronic medical record. At 34 weeks PMA, one buccal swab sample from the infant was obtained. We assessed EFS, in person, at the bedside, once a week, starting after the first PO until discharge. Either the PI or speech therapist on the research team fed infants following the study site's standard feeding protocol (eg, 20–30 minutes, side-lying, swaddling, and external pacing).

Statistical Analysis

Exploratory data analysis was conducted to describe infant characteristics, early life stress, EFS, and DNA methylation of *NR3C1* exon 1F and *HSD11B2* promoter regions. Daily acute stress and chronic stress, as well as their distribution over 60 days of NICU hospitalization, was computed using graphical techniques and summary statistics (mean $\pm$ *SD*). The number of EFS assessments varied between infants; however, all infants had the first EFS evaluation. Thus, data from the first EFS evaluation were used for all bivariate analyses. Scores of acute and chronic stress over entire NICU hospitalization were used for all bivariate analyses. Pearson correlation was used to examine the relationships among EFS subscales. Pearson correlation was computed to assess the relationship between acute/chronic stress over entire NICU hospitalization and EFS and duration to transition from first to full PO. Spearman's rho correlations were computed to assess relationships between DNA methylation of *NR3C1* exon 1F and *HSD11B2* promoter regions and infant characteristics, acute/chronic stress, and EFS. The bivariate analysis was conducted on the percent DNA methylation of each CpG site, as well as the mean percent DNA methylation across all CpG sites. Of note, given the exploratory nature of the analysis, no correction for type 1 error was used, hence caution must be exercised when interpreting the findings.

RESULTS

Feasibility of Study Protocol

We were readily able to screen each infant's medical record upon admission to the NICU. Parents were approached in the postpartum unit as soon as the second day after birth. On average, the study site admits approximately 43 infants who met inclusion criteria per year. Within 6 months of recruitment, we approached and recruited 85% (18) of parents whose infants were eligible and enrolled 65% (11) of these infants. We were unable to meet with some parents due to early postpartum discharge. We could not recruit parents who only spoke Spanish because we did not have access to a translator or Spanish-speaking research staff for consenting. Daily early life stress data from the infants' electronic medical records were extracted as planned. Buccal samples at 34 weeks PMA and weekly EFS assessments were obtained as planned without any adverse effects. We learned that early morning is the best time to collect samples as the unit was often quiet, thus we could easily coordinate the sample collection with the nurses' workflow. We avoided conducting EFS on the day of eye exam since it may affect PO performance. All infants enrolled in the pilot study were bottle-fed. Four of the infants only had one EFS evaluation because they achieved full PO within a week and were discharged before we could conduct the subsequent assessment. For infants who were born at 34 weeks GA and began PO shortly after enrollment, we were able to monitor their PO status closely and coordinate our team's schedule to obtain at least 1 EFS assessment prior to discharge.

Infant Characteristics

We enrolled 11 infants. One infant was withdrawn due to gastrointestinal defects and chromosomal abnormalities diagnosed after enrollment. The final sample size was $N = 10$. Descriptive statistics for infant characteristics are shown in Table 1.

Early Life Stress

Infants experienced 716.41 (mean) ± 478.80 (*SD*) acute stress events over the entire NICU hospitalization, with a daily mean of 19.03 ± 8.14 . Infants experienced a mean of 1644.80 ± 799.27 hours of chronic stress events over the entire NICU hospitalization, with a daily mean of 47.96 ± 30.44 hours. Of note, since some infants encountered multiple procedures classified as chronic stress events, the daily duration (in hours) exceeded 24 hours. Figure 1 shows the change in daily mean frequency of acute and chronic stress events delineated by pain severity levels from 2 to 5 over 60 postnatal days in the NICU. Overall, there was no change in the daily frequency of acute or chronic events over time. However, there were more severe or painful events during the first week of NICU hospitalization. More frequently occurring acute stress events were those that are classified as level 3, eg, diaper changes, position changes, being weighed, and continuous positive airway pressure manipulation. More frequently occurring chronic stress events were those that are classified as level 2, eg, IV in situ, peripherally inserted central catheter line in situ, umbilical venous catheter line in situ, and umbilical artery catheter line in situ.

Oral Feeding Skill Development

All infants had at least 1 EFS assessment, ranging from 1 to 4 assessments. The number of EFS assessments varied between infants due to their individual PO progression and NICU discharge. A total of 18 assessments were completed for all infants. Descriptive statistics for each subscale at the first assessment are shown in Table 2. Exploratory analysis revealed that a higher respiratory regulation score was significantly associated with a higher physiologic stability score ($r = 0.52$, $P = .028$) and a higher oral-motor function score ($r = 0.49$, $P = .035$). A higher physiologic stability score and higher oral-motor function score were also associated with a higher engagement score ($r = 0.61$, $P = .007$ and $r = 0.528$, $P = .024$, respectively). The engagement score was not associated with the respiratory regulation score. The swallowing coordination score was not related to any of the other subscales.

Early Life Stress and Oral Feeding Skill Development

Higher frequency of acute stress events was associated with a lower engagement score ($r = -0.70$, $P = .034$), physiologic stability score ($r = -0.80$, $P = .009$), and longer duration to transition from first to full PO ($r = 0.72$, $P = .018$). The frequency of acute stress events was not associated with respiratory regulation, oral-motor function, or swallowing coordination scores. No significant associations were found between the number of hours of chronic stress events and any EFS subscales.

DNA Methylation of *NR3C1* Exon 1F and *HSD11B2* Promoter Regions

Descriptive statistics for DNA methylation at each CpG site of *NR3C1* exon 1F and *HSD11B2* promoter regions are shown in Supplement Digital Content, Tables 1 and 2, available at: <http://links.lww.com/ANC/A294>, respectively. Significant associations between infant characteristics and DNA methylation of *NR3C1* exon 1F and *HSD11B2* are shown in Tables 3 and 4, respectively.

DNA Methylation of *NR3C1* Exon 1F, Early Life Stress, and Oral Feeding Skills Development

Higher DNA methylation at CpG 25 was associated with higher swallowing coordination score ($\rho = 0.72$, $P = .044$) (Figure 2). Higher DNA methylation at CpG 17 and CpG 31 was associated with higher frequency of acute stress events ($\rho = 0.84$, $P = .004$ and $\rho = 0.85$, $P = .004$, respectively), higher number of hours of chronic stress events ($\rho = 0.74$, $P = .024$ and $\rho = 0.72$, $P = .028$, respectively), but lower engagement score ($\rho = -0.88$, $P = .004$ and $\rho = -0.89$, $P = .003$, respectively), lower physiologic stability score ($\rho = -0.94$, $P < .001$ and $\rho = -0.94$, $P < .001$, respectively), and longer duration to transition from first to full PO ($\rho = 0.75$, $P = .019$ and $\rho = 0.72$, $P = .027$, respectively).

DNA Methylation of *HSD11B2*, Early Life Stress, and Oral Feeding Skills Development

Lower DNA methylation at CpG 21 was associated with lower respiratory regulation score ($\rho = 0.77$, $P = .016$) (Figure 2). Higher DNA methylation at CpG 14 was associated with lower respiratory regulation score ($\rho = -0.76$, $P = .017$) and longer duration to transition from first to full PO ($\rho = 0.79$, $P = .006$). Higher DNA methylation at CpG 4 was associated with higher number of hours of chronic stress events ($\rho = 0.68$, $P = .029$), lower oral-motor

function score ($\rho = -0.67$, $P = .045$), lower engagement score ($\rho = -0.82$, $P = .007$), and longer duration to transition from first to full PO ($\rho = 0.69$, $P = .026$). Higher DNA methylation at CpG 28 was associated with higher frequency of acute stress events ($\rho = 0.84$, $P = .002$) and higher number of hours of chronic stress events ($\rho = 0.64$, $P = .047$), lower engagement score ($\rho = -0.86$, $P = .003$), lower physiologic stability ($\rho = -0.86$, $P = .003$), and longer duration to transition from first to full PO ($\rho = 0.73$, $P = .018$).

DISCUSSION

Overall, the pilot protocol informed planning for future studies. We will continue to integrate the protocol and make appropriate improvements based on the lessons that we learned and the limitations we identified. Research involving families of preterm neonates or neonates in the NICU is critical for improving neonatal care and outcomes. However, recruiting these families poses unique challenges. As observed in our pilot and the literature, barriers to research participation in this population include parents' emotional and psychological stress, limited knowledge about the importance and benefits of research, trust and concerns about privacy, and cultural and language barriers.^{29–31} These barriers can hinder research participation of preterm infants and their families. It is critical to address these barriers through targeted strategies in order for researchers to improve recruitment and retention of these families.

In this study, we detected a low percent of DNA methylation of *NR3C1* exon 1F (0.11%–1.19%) and *HSD11B2* (0.12%–1.00%) promoter regions using buccal cells collected at 34 weeks PMA. Similarly, Lester et al found low percent DNA methylation in buccal cells collected at 38 to 40 weeks PMA, with levels of 0.20% to 0.60% methylation for *NR3C1* exon 1F and 0.60% to 4.00% for *HSD11B2*.⁷ Likewise, Kantake et al reported low blood cell DNA methylation (0.10%–1.60%) for *NR3C1* exon 1F (at birth and 1 month and 2 months after birth).³² As well, using a mix of blood and buccal cells, Chalfun et al reported low DNA methylation (0.28%–1.94%) of *NR3C1* exon 1F in preterm infants at birth and day 5, 30, and 90 of life.¹² However, placental cell DNA methylation was reported to be higher (1.70%–3.90%) for *NR3C1* exon 1F and (1.50%–4.90%) for *HSD11B2* promoter.¹⁰ We acknowledge that there is a potential for tissue specificity of DNA methylation.³³ However, the percent of DNA methylation of *NR3C1* exon 1F and *HSD11B2* promoter regions reported in the literature is consistently in the low range across cell types and infant age at the time of sample collection. Whether such low DNA methylation is biologically meaningful or sufficient to alter phenotypes remains unclear.^{12,34} In a review paper, Leenen et al noted that low DNA methylation of *NR3C1* elicited associated transcription, protein synthesis, and phenotype changes and have been linked with many complex diseases such as type 2 diabetes, major depression, schizophrenia, hypertension, and cardiovascular diseases.³⁴ Understanding the biological significance and meaning of low DNA methylation awaits further research.

To our knowledge, this pilot study is the first to find significant associations among early life stress, DNA methylation of *NR3C1* exon 1F and *HSD11B2* promoter regions, and PO skill development in preterm infants. Infants who experienced more acute and chronic stress during NICU hospitalization exhibited higher DNA methylation at CpG 17 and CpG

31 of the *NR3C1* exon 1F, as well as CpG 4 and CpG 28 of the *HSD11B2* promoter regions. Infants with higher DNA methylation at these CpG sites or infants who experienced more acute stress events during NICU hospitalization also exhibited less optimal PO skill development and experienced longer transition from first to full PO. Future studies with larger sample sizes are needed to test whether DNA methylation of these CpG sites mediates the association of early life stress on PO skill development.

Our preliminary findings are novel and extend the limited evidence in the literature regarding early life stress and DNA methylation in infants. Prior studies suggest that altered DNA methylation of *NR3C1* and *HSD11B2* genes could be the biomechanism responsible for poor neurobehavioral development.^{6,11,35–38} For example, in full-term infants, placental cell DNA methylation of CpG 2 of the *NR3C1* exon 1F and CpG 4 of the *HSD11B2* promoter was related to poor behavioral self-regulation, low muscle tone, and lethargy.³⁶ Lower placental cell DNA methylation of CpG 6, 7, 8, and 10 of the *NR3C1* exon 1F was related to higher quality of movement and attention.³⁵ Lower placental cell DNA methylation of *HSD11B2* promoter was related to higher quality of movement.³⁸ Infants with low placental cell DNA methylation of *NR3C1* exon 1F and high *HSD11B2* exhibited lower excitability, while infants with high *NR3C1* exon 1F and low *HSD11B2* exhibited higher asymmetrical reflexes.¹¹ Similarly, in buccal cells, high-risk infants had higher buccal cell DNA methylation at the CpG 3 of the *NR3C1* exon 1F and lower DNA methylation at CpG 3 of the *HSD11B2* promoter compared to low-risk infants.⁷ As well, critical reviews confirmed an association between early life stress and higher DNA methylation of *NR3C1* exon 1F in cord blood, peripheral blood, buccal, and placental cells.^{39,40} Although there is some overlapping in DNA methylation in specific CpG sites, the associative findings are scattered across many different CpG sites within the *NR3C1* exon 1F or *HSD11B2* promoter regions. This is not surprising given the different tissues and developmental windows of time at sample collection and the complexity of the DNA methylation process. The data generated thus far represent an important first step in elucidating epigenetic processes that mediate environmental influences on development. Future studies are necessary to investigate the complex interactions between the different genes and CpG sites involved in transcription.

Taken together, the findings from the literature suggest that preterm infants' postnatal exposure to an adverse environment or early life stress influences cortisol regulation via DNA methylation of *NR3C1* exon 1F and *HSD11B2* promoter regions, which can, in turn, alter neurobehavioral development. The integrity of neurobehavioral development supports the infant to engage during feeding, organize oral-motor movements, coordinate sucking-breathing-swallowing, and maintain physiologic stability, allowing for oral feeding safety and efficiency.^{19,26} Our preliminary findings showing significant associations among early life stress, DNA methylation of *NR3C1* exon 1F and *HSD11B2* promoter regions, and PO skill development in preterm infants supports an epigenetic mechanism linking early life stress, neurobehavioral development, and oral feeding skill development.

Additionally, we observed several associations between infants' health status and DNA methylation of *NR3C1* exon 1F and *HSD11B2* promoter regions. For example, infants who were born at younger GA, and had lower birth weight and Apgar scores exhibited changes

in DNA methylations at multiple CpG sites of the *NR3C1* exon 1F and *HSD11B2* promoter. Oxygen and energy homeostasis from stable cardiopulmonary function is necessary for epigenetic programming by DNA methylation.¹² Given this, Chalfun et al hypothesized that a low Apgar score, which suggests adverse cardiopulmonary function, may inactivate the enzymatic cascade that leads to changes in DNA methylation.¹² Additionally, we found that infants who had more serious health complications during their NICU hospitalization demonstrated higher DNA methylation at CpG 25 of the *NR3C1* exon 1F and CpG 4 of the *HSD11B2* promoter regions. Previously, Kantake et al found associations between DNA methylation of *NR3C1* exon 1F promoter regions at CpG 4, 8, 10, 25, and 29 and GA, intrauterine growth, Apgar score, mode of delivery, and NICU admission.⁴¹ Others found that infants with several medical morbidities exhibited lower DNA methylation at the CpG 1 site of the *NR3C1* exon 1F promoter region.⁴² Health morbidities likely engender a biologic stress response, leading to altered DNA methylation of CpG sites within the exon 1F of *NR3C1* and *HSD11B2* promoter regions.

Not surprisingly, differences in DNA methylation patterns of the *NR3C1* exon 1F promoter region are observed between full-term vs preterm infants and over time after birth,^{12,41} which likely reflects biological maturity. However, it may also be related to early life stress exposure associated with preterm birth and NICU admission.¹² Concurring with our findings, evidence confirms that premature birth and its associated birth-related outcomes, as well as health complications during NICU hospitalization, may be considered risk factors for negative epigenetic modifications. Collectively, current scientific evidence supports the hypothesis that prematurity and adverse experiences during NICU hospitalization would result in altered DNA methylation of *NR3C1* exon 1F, leading to HPA dysregulation and altered cortisol levels in preterm infants.

The strengths of this pilot study allowed for the development of a workable and scalable protocol. Logistic challenges and resolutions were identified while working closely with the NICU clinical team. Successful recruitment approaches and data collection methods were refined. Importantly, the data obtained will aid in estimating sample sizes for future studies investigating epigenetic mechanisms linking early life stress and PO skill development.

Generalization of the reported findings is limited by the exploratory nature of this pilot study design and the small sample size. We did not conduct a power analysis. Thus, the study may be underpowered to evaluate EFS outcomes and to control for all variables. The measure of acute and chronic stress events was primarily based on medical record review. Although the NISS is a well-developed instrument, the NISS score is still considered an indirect measure of stress because health professionals perceive the level of each stressful event in the NICU. Additionally, we evaluated the EFS in person at the bedside. This approach can increase the risk of subjective bias, compared to other objective methods, such as videofluoroscopic swallow study. However, to ensure inter-rater reliability, the PI and the speech therapist double-coded 30% of the in-person EFS assessments. At the conclusion of the assessments, they discussed any discrepancies until 100% agreement was achieved. The sample had a wide range of baseline GAs, which could affect stress exposure and PO skill development over the NICU hospitalization. Thus, it is possible that GA may confound the study results. However, due to the small sample size and the bivariate analysis

nature of this pilot, we did not control this potential confounding variable. Similarly, the findings focused on individual CpG site within a single gene promoter. However, there are multiple CpG sites and several other genes related to the HPA axis, cortisol regulation, and the stress response (eg, *NFKB1A*, *FKBP5*, *BDNF*, *GILZ*, and *SLC6A4*). Complex interactions between the genes and CpG sites within each gene pose major challenges in the interpretations and implications of study results. In spite of these limitations, this pilot serves as a guide for future well-powered studies aimed at understanding the underlying biobehavioral epigenetics of early life stress on PO skills.

IMPLICATIONS FOR PRACTICE AND RESEARCH

Descriptive statistics of early life stress from this pilot study inform nursing practice. It may be helpful for neonatal nurses to see tangible quantification of stressors related to NICU care and environment, thus perhaps aiding intention and effort to reduce unnecessary stressors. Preliminary evidence supports the hypothesis that early life stress and health status at birth and during NICU hospitalization may contribute to DNA methylation of *NR3C1* and *HSD11B2*. Prior work suggests that stress-related negative epigenetic modifications are stable and can lead to detrimental short-term and long-term outcomes.^{2,4,43,44} These findings prompt neonatal nurses to think about how critical it is to ameliorate these negative consequences, by enriching the NICU environment, reducing acute and chronic stress events, and enhancing infant protective mechanisms. We acknowledge that premature birth and NICU hospitalization are inevitably associated with early life stress due to early and prolonged parental separation, sleep disturbances, excessive environmental stimulation, discomfort from routine care procedures, and painful stimulation from life-saving medical procedures.^{45–49} As well, identification of the biobehavioral epigenetic mechanisms may positively impact nursing practice and research. Epigenetic modifications are sensitive to environmental and life experiences.⁸ Perhaps, a pivotal nursing goal is to promote positive epigenetic modifications and minimize/reverse the negative epigenetic modifications. Prior animal studies demonstrate that early exposure to more licking and grooming is capable of reversing early stress-related DNA methylation of glucocorticoid-related genes.^{2,3,50} In human infants, lactation also reversed DNA methylation of glucocorticoid-related genes.⁵⁰ Future studies may consider accounting for variables such as lactation and rooming in, as each may attenuate the impact of stress-related DNA methylation. Investigation of early life stress and strategies to reduce its impact can build a foundation for elevating neonatal nursing care and improved infant outcomes.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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What This Study Adds

- The findings support a potential biobehavioral epigenetic mechanism linking early life stress and oral feeding skill development.
- Early life stress and health status at birth and during NICU hospitalization may contribute to DNA methylation of *NR3C1* and *HSD11B2* promoter regions.
- Health morbidities likely engender a biological stress response, leading to altered DNA methylation of CpG sites within the exon 1F of *NR3C1* and *HSD11B2* promoter regions.

Summary of Recommendations

What we know:

- Preterm infants experience early life stress due to the NICU environment, health conditions, caregiving interventions, and prolonged parental separation.
- Early life adversities and environmental stimuli modify DNA methylation of 2 glucocorticoid-related genes, namely *NR3C1* and *HSD11B2*.
- DNA methylation of *NR3C1* and *HSD11B2* may alter HPA axis regulation of cortisol and interfere with neurobehavioral development.
- The integrity of neurobehaviors is crucial for oral feeding skill development.

What needs to be studied:

- Further understanding of the relationship between early life stress and DNA methylation of *NR3C1* and *HSD11B2*.
- The relationship between early life stress and oral feeding skill development.
- The impact of DNA methylation of *NR3C1* and *HSD11B2* on oral feeding skill development.
- The multiomics pathways whereby early life stress influences the phenotypes of infant outcomes.
- The impact of prenatal factors as well as parents/caregivers' stressors and related social determinants of health and how they can buffer early life stress.

What can we do today:

- Enriching the NICU environment, reducing acute and chronic stress events, and enhancing infant protective mechanisms are imperative.
- Promoting positive epigenetic modifications and minimizing the negative epigenetic modifications.
- Advocating and implementing care practices, eg, skin-to-skin care, lactation, developmental care, trauma-informed care, to facilitate nurturing and enjoyable experiences to minimize early life stress and support infant's growth and development.

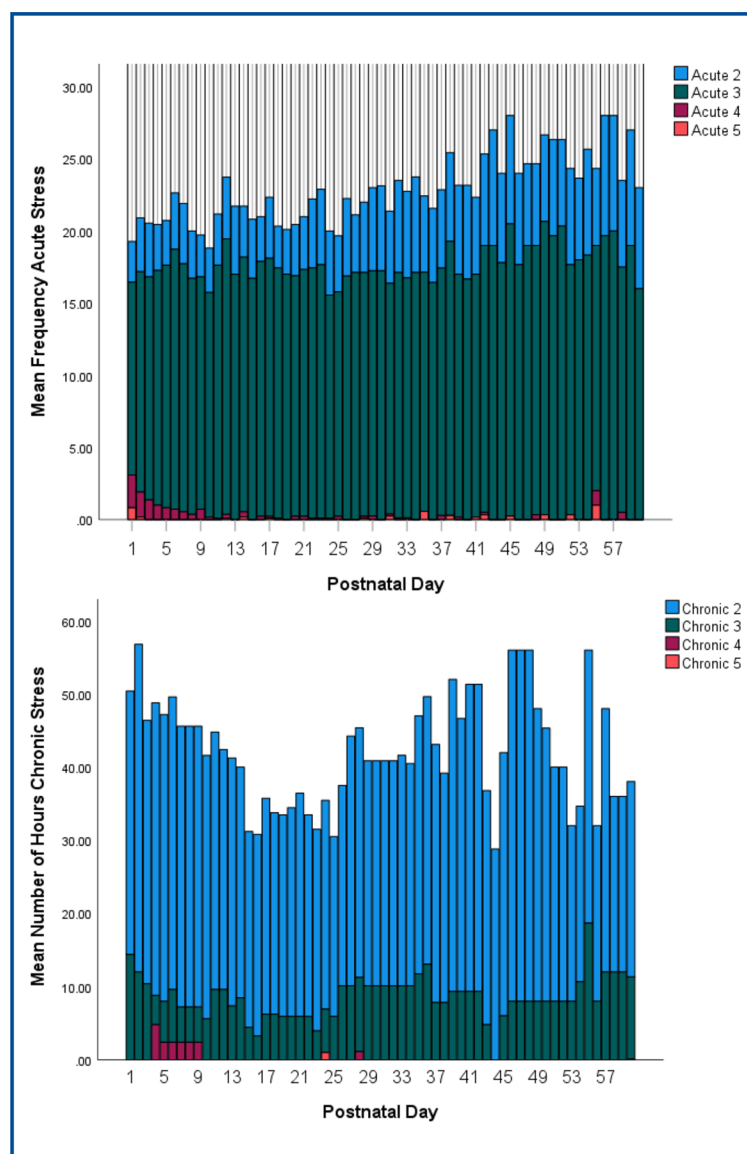
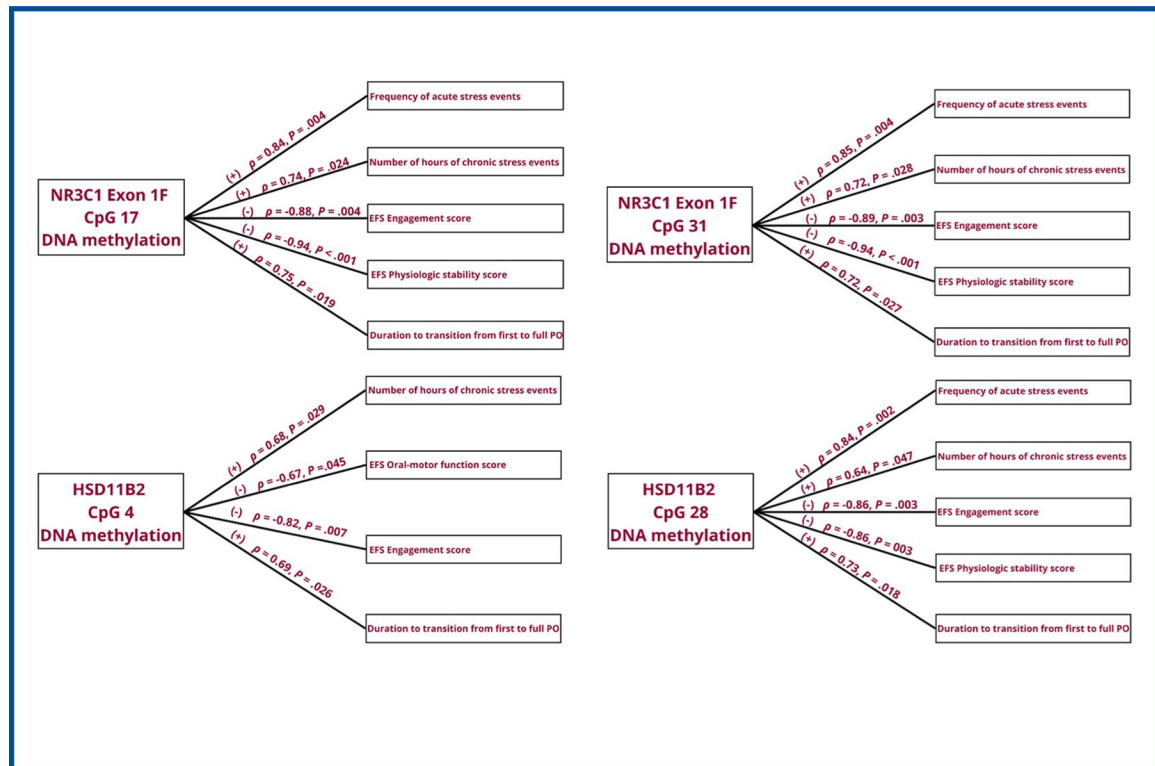


FIGURE 1. Mean frequency of daily early life stress during the first 60 postnatal days delineated by severity levels (N = 10).

**FIGURE 2.**

The relationships between DNA methylation of *NR3C1* and *HSD11B2*, early life stress, and oral feeding skill development ($N = 10$). The significant associations between DNA methylation of 4 specific CpG sites of the 2 candidate genes, frequency of acute stress events, number of hours of chronic stress events, and subscales of EFS are illustrated in Figure 2. Negative associations are noted with (–) and positive associations are noted with (+). EFS indicates Early Feeding Skills Assessment.

TABLE 1.

Descriptive Statistics for Infant Characteristics (N = 10)

Infant Characteristics	Percent (%)	n
Race		
Asian	10	1
Black or African American	10	1
White	80	8
Ethnicity		
Not Hispanic or Latino	30	3
Hispanic or Latino	70	7
Infant sex		
Male	50	5
Female	50	5
Neonatal Medical Index Classification		
Classification 1	0	0
Classification 2	60	6
Classification 3	30	3
Classification 4	10	1
Classification 5	0	0
Respiratory distress syndrome		
Yes	90	9
No	10	1
Small for gestational age		
Yes	10	1
No	90	9

Infant Characteristics	Mean	SD	Median (IQR)	Range
Gestational age, weeks	30.40	2.55	30.50 (3.75)	26.00–34.00
Birth weight, g	1478.50	500.00	1322.50 (1070.00)	970.00–2260.00
1-minute Apgar score	7.40	1.57	8.00 (2.25)	4.00–9.00
5-minute Apgar score	8.80	0.42	9.00 (0.25)	8.00–9.00

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Infant Characteristics	Percent (%)	n	
Duration of exclusive tube feeding, days	16.90	12.69	16.50 (21.00) 0.00–40.00
Duration of tube feeding, days	33.60	17.17	33.00 (35.00) 11.00–58.00
Duration of transition from first to full PO, days	17.60	7.80	13.50 (10.50) 12.00–35.00
Duration of NICU hospitalization, days	41.90	22.71	40.50 (34.25) 15.00–90.00
PMA at first PO, weeks	33.66	0.72	33.79 (0.75) 31.80–34.00
PMA at full PO, weeks	36.40	1.50	36.36 (2.18) 33.70–38.40
PMA at discharge, weeks	36.97	1.44	36.350 (2.36) 34.70–39.30

Abbreviations: IQR, interquartile range; NICU, neonatal intensive care unit; PMA, postmenstrual age; PO, oral feeding.

TABLE 2.
Descriptive Statistics for Early Feeding Skill Assessment at the First Assessment (N = 10)

Early Feeding Skills Assessment Subscale	Mean	SD	Median (IQR)	Range
Respiratory Regulation	9.70	2.01	10.00 (3.00)	6.00–12.00
Oral-Motor Function	8.50	1.90	9.00 (4.00)	6.00–11.00
Swallowing Coordination	11.00	1.05	11.00 (2.00)	9.00–12.00
Engagement	4.40	0.97	5.00 (1.00)	2.00–5.00
Physiologic Stability	10.50	1.43	11.00 (1.00)	7.00–12.00

Abbreviation: IQR, interquartile range.

TABLE 3.DNA Methylation of *NR3C1* Exon 1F and Infant Characteristics (N = 10)

Infant Characteristics	CpG Site	ρ	P
Gestational age	26	0.68	.045
Birth weight	15	0.67	.048
	30	0.68	.044
1-minute Apgar score	14	0.68	.042
	3	-0.74	.024
	11	-0.70	.035
	20	-0.74	.022
	25	-0.75	.019
5-minute Apgar score	24	-0.71	.033
Neonatal Medical Index Classification	25	0.81	.009

TABLE 4.DNA Methylation of *HSD11B2* and Infant Characteristics (N = 10)

Infant Characteristics	CpG Site	ρ	P
Gestational age	14	-0.65	.043
Birth weight	21	0.65	.041
1-minute Apgar score	17	0.65	.042
	2	-0.74	.014
	3	-0.66	.038
	5	-0.79	.007
	16	-0.79	.006
5-minute Apgar score	2	-0.66	.039
	16	-0.65	.043
Neonatal Medical Index Classification	4	0.69	.026